

A Practical Guide to Instruction tuning

TOPIC -- INSTRUCTION TUNING

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Denial of service due to scraping Web scraping is used to gather training data for LLMs. This produces large volumes of traffic which has led to denial-of-service issues with many websites. The situation has been described as "a DDoS on the entire internet" and in some cases scrapers make up the majority of traffic to a site. AI web crawlers may bypass the methods that are usually used to block web scrapers, such as robots.txt files, blocking user-agents and filtering suspicious traffic. Website operators have resorted to novel methods such as AI tarpits, but some fear that tarpits will only worsen the burden on servers.

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reported arithmetics decoding the International Phonetic Alphabet unscrambling a word's letters disambiguating word-in-context datasets converting spatial words cardinal directions (for example, replying "northeast" in response to a 3x3 grid of 8 zeros and a 1 in the top-right), color terms represented in text. chain-of-thought prompting: In a 2022 research paper, chain-of-thought prompting only improved the performance for models that had at least 62B parameters. Smaller models perform better when prompted to answer immediately, without chain of thought. identifying offensive content in paragraphs of Hinglish (a combination of Hindi and English), and generating a similar English equivalent of Kiswahili proverbs. Schaeffer et al. argue that the emergent abilities are not unpredictably acquired, but predictably acquired according to a smooth scaling law. The authors considered a toy statistical model of an LLM solving multiple-choice questions, and showed that this statistical model, modified to account for other types of tasks, applies to these tasks as well. Let x be the number of parameter count, and y be the performance of the model.

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Selection bias Selection bias refers the inherent tendency of large language models to favor certain option identifiers irrespective of the actual content of the options. This bias primarily stems from token bias—that is, the model assigns a higher a priori probability to specific answer tokens (such as "A") when generating responses. As a result, when the ordering of options is altered (for example, by systematically moving the correct answer to different positions), the

model's performance can fluctuate significantly. This phenomenon undermines the reliability of large language models in multiple-choice settings.

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NLP researchers were evenly split when asked, in a 2022 survey, whether (untuned) LLMs "could (ever) understand natural language in some nontrivial sense". Proponents of "LLM understanding" believe that some LLM abilities, such as mathematical reasoning, imply an ability to "understand" certain concepts. A Microsoft team argued in 2023 that GPT-4 "can solve novel and difficult tasks that span mathematics, coding, vision, medicine, law, psychology and more" and that GPT-4 "could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence system": "Can one reasonably say that a system that passes exams for software engineering candidates is not really intelligent?" Ilya Sutskever argues that predicting the next word sometimes involves reasoning and deep insights, for example if the LLM has to predict the name of the criminal in an unknown detective novel after processing the entire story leading up to the revelation. Some researchers characterize LLMs as "alien intelligence". For example, Conjecture CEO Connor Leahy considers untuned LLMs to be like inscrutable alien "Shoggoths", and believes that RLHF tuning creates a "smiling facade" obscuring the inner workings of the LLM: "If you don't push it too far, the smiley face stays on. But then you give it [an unexpected] prompt, and suddenly you see this massive underbelly of insanity, of weird thought processes and clearly non-human understanding." In contrast, some skeptics of LLM understanding believe that existing LLMs are "simply remixing and recombining existing writing", a phenomenon known as stochastic parrot, or they point to the deficits existing LLMs continue to have in prediction skills, reasoning skills, agency, and explainability. For example, GPT-4 has natural deficits in planning and in real-time learning. Generative LLMs have been observed to confidently assert claims of fact which do not seem to be justified by their training data, a phenomenon which has been termed "hallucination". Specifically, hallucinations in the context of LLMs correspond to the generation of text or responses that seem syntactically sound, fluent, and natural but are factually incorrect, nonsensical, or unfaithful to the provided source input. Neuroscientist Terrence Sejnowski has argued that "The diverging opinions of experts on the intelligence of LLMs suggests that our old ideas based on natural intelligence are inadequate". Efforts to reduce or compensate for hallucinations have employed automated reasoning, retrieval-augmented generation (RAG), fine-tuning, and other methods. The matter of LLM's exhibiting intelligence or understanding has two main aspects—the first is how to model thought and language in a computer system, and the second is how to enable the computer system to generate

human-like language. These aspects of language as a model of cognition have been developed in the field of cognitive linguistics. American linguist George Lakoff presented neural theory of language (NTL) as a computational basis for using language as a model of learning tasks and understanding. The NTL model outlines how specific neural structures of the human brain shape the nature of thought and language and in turn what are the computational properties of such neural systems that can be applied to model thought and language in a computer system. After a framework for modeling language in a computer systems was established, the focus shifted to establishing frameworks for computer systems to generate language with acceptable grammar. In his 2014 book titled *The Language Myth: Why Language Is Not An Instinct*, British cognitive linguist and digital communication technologist Vyvyan Evans mapped out the role of probabilistic context-free grammar (PCFG) in enabling NLP to model cognitive patterns and generate human-like language.

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Sentience Contemporary AI practitioners generally agree that present-day large language models do not exhibit sentience. A minority view argues that even if there is a small chance that a given software system can have subjective experience, which some philosophers suggest is possible, then ethical considerations around potential large-scale suffering in AI systems may need to be taken seriously—similar to considerations given to animal welfare. Proponents of this view have proposed various precautionary measures like moratoriums on AI development and induced amnesia to address these ethical concerns. Leonard Dung argues that the evidential frameworks used to assess consciousness in animals apply equally to AI systems and that there is a significant probability near-future AI will be capable of suffering, making AI suffering risk a serious near-term ethical concern that requires systematic mitigation. On the other hand, some existential philosophers argue there is no generally accepted way to determine if an LLM is conscious, given the inherent difficulty of measuring subjective experience. The 2022 Google LaMDA incident, where engineer Blake Lemoine claimed that the model was conscious, highlighted how LLMs can convince users that they are sentient through responses that do not prove sentience. Google described the engineer's claims as unfounded, and he was dismissed. Murray Shanahan argues that anthropomorphic framing of LLM capabilities encourages unwarranted attribution of cognitive properties to systems that operate through statistical pattern completion. Kristina Šekrst develops this further, arguing that LLMs function as "illusion engines" capable of producing outputs that coherently simulate properties such as consciousness without possessing them, but highlighting that, due to sophisticated creativity-temperature tradeoff, we may never be certain

whether we are dealing with the emergence of consciousness or just a hallucination. David Chalmers similarly argues that while current LLMs likely lack features considered necessary for consciousness, extended successors incorporating these elements could plausibly meet the criteria within a decade.